

Course Code:		Course Title - Quantum-Inspired Computational Methods for AI and Computer Vision	
School/Centre:		School of Computer Science and Engineering [SCOPE]	
Beneficiaries: UG/PG/Research		UG/PG/Research	
Course Objectives		1. Provide a foundational understanding of quantum-inspired computational concepts and their relevance to Artificial Intelligence (AI) and Computer Vision. 2. Introduce AI-driven image processing and computer vision techniques for intelligent visual analytics and pattern recognition tasks. 3. Equip learners with practical understanding of modelling and simulation-oriented approaches supporting analysis, monitoring, and decision-support applications.	
Course Outcomes		1. Develop basic AI and Computer Vision models for image analysis, recognition, and visual inference tasks. 2. Evaluate AI and image processing techniques for solving real-world visual analytics and automation problems. 3. Analyze practical challenges in deploying intelligent vision systems, including reliability, interpretability, scalability, and ethical considerations.	
S. No.	Topic		Duration (Hours)
1.	Introduction to Quantum-Inspired Computing Concepts Conceptual Foundations: Motivation for Quantum-Inspired Methods, Classical vs Quantum Computing (high-level view), Why “quantum-inspired” ≠ quantum hardware, Relevance to AI & optimization problems Tools/libraries: Qiskit, PennyLane, QuTiP		6
2.	Mathematical & Representational Intuition: Vector space interpretation, Probability amplitudes analogy, Superposition as modelling inspiration, Basic idea of quantum state encoding Tools/libraries: NumPy, TensorNetwork, SymPy		6
3.	Quantum-Inspired Representations in AI & Vision: Encoding classical data into abstract state representations, Feature space transformations and similarity concepts, Connection to pattern recognition and Computer Vision, High-dimensional representations in image analysis Tools/libraries: scikit-learn, Faiss		6
4.	Applications in AI and Computer Vision: Quantum-inspired optimisation intuition, Quantum-inspired kernels / similarity measures, Emerging applications in AI and Computer Vision tasks, Conceptual examples in image classification & recognition Tools/libraries: Ocean SDK, NEAL, DEAP		6
5.	Practical Considerations & Limitations: Computational feasibility & realism, Misconceptions about quantum advantage, Limitations & realistic expectations Research opportunities & future directions		6

	Tools/libraries: Profilers, Benchmarking	
	Mode of Evaluation: Quiz - 30%, Mini-Project-40% and Viva Voce -30%	
	Total Course Duration	30 hours
	Date of Board of Studies in which it is recommended	

Books:

1. Jadhav, D., Wani, P., Dasre, N., & Mallik, B. B. (Eds.). (2026). Quantum-Inspired Approaches for Intelligent Data Processing. John Wiley & Sons.
2. Torralba, A., Isola, P., & Freeman, W. T. (2024). *Foundations of computer vision*. MIT Press.
3. Zhang, X., Mao, Z., Chimitt, N., & Chan, S. H. (2024). Imaging through the atmosphere using turbulence mitigation transformer. *IEEE Transactions on Computational Imaging*, 10, 115-128.

Course Abstract:

This course provides an introduction to quantum-inspired computational concepts and their relevance to Artificial Intelligence (AI) and Computer Vision, focusing on how ideas such as high-dimensional representations, vector space modelling, and probabilistic interpretations can enhance classical data analysis and intelligent systems. Rather than emphasizing quantum hardware or advanced quantum mechanics, the course highlights conceptual understanding and practical intuition behind quantum-inspired techniques that support modern AI and vision applications, including feature representation, similarity analysis, and interpretable modelling. Designed as a Value Added Programme (VAP), the course promotes interdisciplinary learning, computational reasoning, and applied analytical thinking, enabling participants to appreciate the capabilities, limitations, and emerging potential of quantum-inspired approaches in AI-driven Computer Vision.

Benchmark Institutions:

Institution	Course/Program	Estimated % of Match
Carnegie Mellon University (CMU) – Pittsburgh, PA, USA	AI / Machine Learning / Computer Vision instructional themes	35
Indian Institute of Science (IISc), Bengaluru give percentage of mapping with this	AI / Computational Modelling / Data-Driven Systems themes	40